Cleaning/ Inspection Procedures for Fan Motor of Spindle Motor

1. Outline

In order to guarantee enough cooling of spindle motors, regular cleaning and inspection is necessary as described in DESCRIPTIONS and MAINTENANCE MANUAL.

However, there are cases in which cleaning and inspection is not conducted regularly and therefore leading to failure and trouble. If the fan motor stop, an overheat alarm may occur and the machine may stop. In the worst case, driving under conditions in which cutting fluid or its mist is splashed on the motor may lead to combustibility of sludge accumulated on the fan motors(*). Regular cleaning and inspection is highly recommended.

Please contact FANUC if there is any question regarding the contents of this report. FANUC service department will conduct inspection at your request so please contact FANUC if necessary.

(*) Our fan cover is made with self-extinguishing material which the flammability is V-0, so if there is no foreign matter, it is not combustibility.

2. Frequency of Cleaning and Inspection

Recommended: once every three months

The standard frequency of cleaning and inspection is once every three months. The frequency may differ according to the condition the motor is used in.

3. Safety Precautions

When cleaning and inspecting the fan motors, please keep in mind the precautions described as follows.

⚠ WARNING

- Before touching a motor, shut off the power to it.

  Even if a motor is not rotating, there may be a voltage across the terminals of the motor.
  Especially before touching a power supply connection, take sufficient precautions.
  Otherwise you may get electric shocks.

- A voltage is applied for a while after the power is shut off. Before touching a motor, be sure to check 0 V.

  There is a high voltage across power terminals for a while after the power is shut off. Do not touch a motor or connect the motor to another device before checking the voltage. Otherwise you may get electric shocks or the motor may get damaged. Be sure to check that the voltage is lowered to 0 V, then touch the motor.

- Do not insert your finger or a stick into the fan motor.

  A cover is mounted on the fan motor, but a stick may get inside it. When the fan is rotating, inserting your finger or a stick may injure you.

⚠ CAUTION

- Do not touch a motor when it is running or immediately after it stops.

  A motor may get hot when it is running. Do not touch the motor before it gets cool enough. Otherwise, you may get burned.

- Be careful not get your hair or cloths caught in a fan.

  Be careful especially for a fan used to generate an inward air flow.

  Be careful also for a fan even when the motor is stopped, because it continues to rotate while the amplifier is turned on.

- Be sure to connect motor cables correctly.

  An incorrect connection of a cable cause abnormal heat generation, equipment malfunction, or failure.

  Always use a cable with an appropriate current carrying capacity (or thickness). For how to connect cables to motors, refer to their respective specification manuals.

※ Precautions are classified into Warning and Caution according to their bearing on safety. Read all Warnings and Cautions thoroughly before attempting to use the machine.

⚠ **WARNING**

Applied when there is a danger of the user being injured or when there is a danger of both the user being injured and the equipment being damaged if the approved procedure is not observed.

⚠ **CAUTION**

Applied when there is a danger of the equipment being damaged, if the approved procedure is not observed.

4. Cleaning/ Inspection Procedures and Work Descriptions

4.1 Cleaning/ Inspection Procedures

Regular inspection is composed of checking mechanical (i.e. rotation) and electrical state of the fan motor.

Follow the procedures below and clean or replace the fan motor in accordance.
As for the details of the content and criteria, refer to Subsection 4.2.

4.1.1 Mechanical Inspection Procedures

If the state of the fan motor is not visible, follow the procedures in Subsection 4.1.2 “Electrical Inspection Procedures, Cleaning and Replacement of the Fan Motor”.

If the state is visible, check the state of rotation.
Check if the fan motor is rotating or not when the power is on and follow the procedures below.

1) If the Fan Motor is Rotating

Go on to Subsection 4.1.2 “Electrical Inspection Procedures, Cleaning and Replacement of the Fan Motor” and follow the procedures.

If the fan motor is clogged with foreign materials such as dust and metal chips, spindle motor’s cooling will be insufficient and may cause failure and trouble. If there is a visible foreign materials, please clean the fan motor following the procedures in Subsection 4.2 “Cleaning of the Fan Motor”.

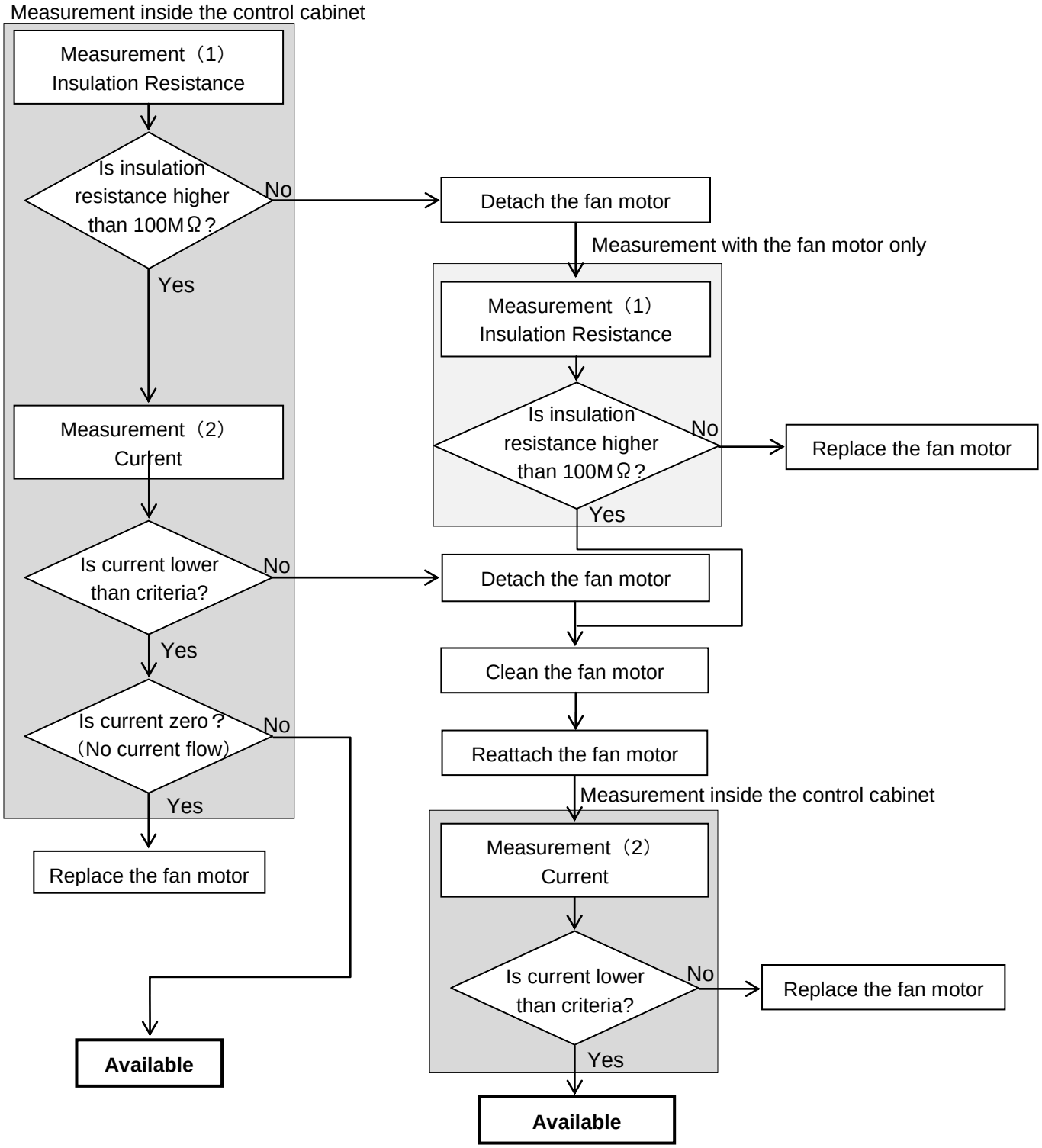
2) If the Fan Motor is not Rotating

Replace the fan motor following the procedure in Subsection 4.2 “Replacement of the Fan Motor”

4.1.2 Electrical Inspection Procedures, Cleaning and Replacement of the Fan Motor

Follow the procedures below and clean or replace the fan motor in accordance.

As for the details of the content and criteria, refer to Subsection 4.2.

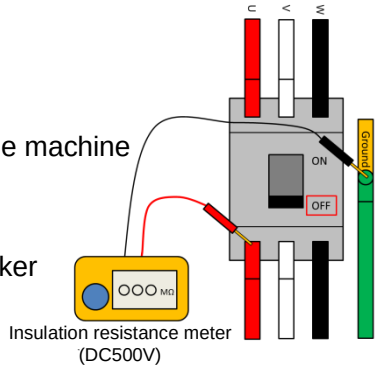


※ Please contact FANUC if there is any question regarding the contents of this report. FANUC service department will conduct inspection at your request so please contact FANUC if necessary.

4.2 Work Descriptions and Criteria for Cleaning and Inspection

※ If you do not own an insulation resistance meter and a clamp meter necessary for the following measurements, confirm that the fan motor is rotating correctly by vision. If the fan motor is found to be not clean, follow the procedures “Cleaning of the Fan Motor”.

Measurement （1） Insulation Resistance



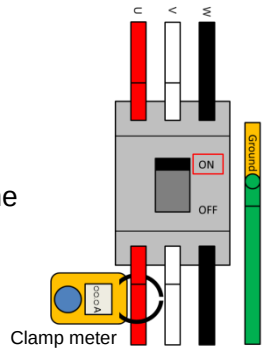
Instrument : insulation resistance meter (DC500V)
Point : breaker or terminal block inside the control cabinet of the machine
Method :

① Turn off the power of the machine.
② Measure the insulation resistance using the terminal on the breaker or terminal block connected to power leads of the fan motor. Measure between one of the three power leads and ground.

Criteria : No failure if the insulation resistance is above 100MΩ.

※ If the fan motor is sharing the same breaker with other equipment, measure the insulation resistance between the power leads of fan motor and the ground wire by isolating the power leads of fan motor or isolating the fan motor.

Measurement（2） Current



Instrument : clamp meter
Point : power leads of fan motor inside the control cabinet of the machine
Method :

① Attach a clamp meter to one of the power leads of the fan motor.
② Turn on the power of the fan motor.
③ Measure the current flowing to the fan motor.

Criteria : No failure if the current is below the criterial value shown in the chart below (i.e. 20% higher than the rated current), and above zero.

⚠ WARNING

- The measurement is done while the power is on. Take enough safety measures to prevent electric shock.

Current Criteria （αiI-B,βiI-B Series）

For model series aside from **αiI-B** and **βiI-B**, contact FANUC sales department or service department.

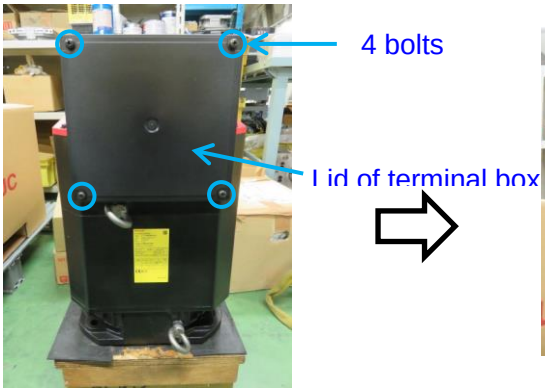
αiI-B, αiIP-B, αiIT-B, αiIPT-B, βiI-B, βiIP-B, βiIT-B Series 200V Type	**50Hz**		**60Hz**	
	Rated Current [A]	**Criterial Value [A]**	**Rated Current [A]**	**Criterial Value [A]**
αiI 1-B, αiI 1.5-B, αiIT 1-B, αiIT 1.5-B	0.09	0.11	0.10	0.12
αiI 2-B, αiI 3-B, αiIT 2-B, αiIT 3-B, βiI 3-B, βiI 6-B, βiIC 3-B, βiIC 6-B	0.09	0.11	0.10	0.12
αiI 6-B, αiI 8-B, αiIT 6-B, αiIT 8-B, βiI 8-B, βiI 12-B, βiIP 12-B, βiIT 12-B, βiIC 8-B	0.13	0.16	0.14	0.17
αiI 12-B～αiI 22-B, αiIP 12-B～αiIP 22-B, αiIT 12-B～αiIT 22-B, αiIPT 22-B, βiI 15-B, βiIP 15-B～βiIP 30-B, βiIT 15-B	0.20	0.24	0.30	0.36
αiI 30-B,αiI 40-B, αiIP 30-B, αiIP 40-B, αiIP 50-B, βiIP 40-B	0.45	0.54	0.48	0.58
αiI 50-B, αiI 60-B, αiIP 60-B	0.42	0.50	0.55	0.66

αiI-B, αiIP-B, αiIT-B, βiI-B, βiIT-B Series 400VType	**50Hz**		**60Hz**	
	Rated Current [A]	**Criterial Value [A]**	**Rated Current [A]**	**Criterial Value [A]**
αiI 1HV-B, αiI 1.5HV-B, αiIT 1.5HV-B	0.09	0.11	0.11	0.13
αiI 2HV-B, αiI 3HV-B, αiIT 2HV-B, αiIT 3HV-B, βiI 3HV-B, βiI 6HV-B,	0.11	0.13	0.13	0.16
αiI 6HV-B, αiI 8HV-B,αiIT 6HV-B, αiIT 8HV-B, βiI 8HV-B, βiI 12HV-B	0.07	0.08	0.08	0.10
αiI 12HV-B～αiI 22HV-B, αiIP 15HV-B, αiIP 22HV-B, αiIT 15HV-B, αiIT 22HV-B, βiI 15HV-B	0.15	0.18	0.16	0.19
αiI 30HV-B, αiI 40HV-B, αiIP 40HV-B, αiIP 50HV-B	0.22	0.26	0.26	0.31
αiI 50HV-B, αiI 60HV-B, αiIP 60HV-B	0.22	0.26	0.28	0.34
αiI 75HV-B, αiI 100HV-B	0.30	0.36	0.36	0.43
αiI 150HV-B	0.70	0.84	1.2	1.44

; αiI, βiI series are similar to the above.

Detaching the Fan Motor

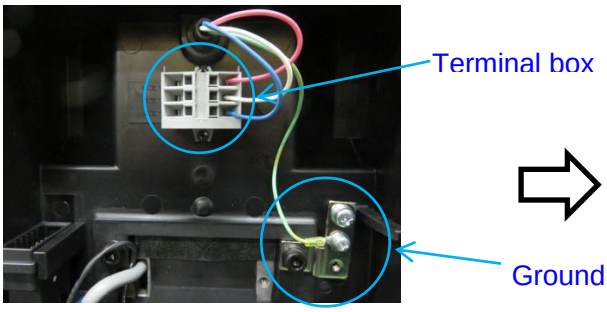
① Unscrew the 4 bolts on the terminal box and remove the lid of terminal box.

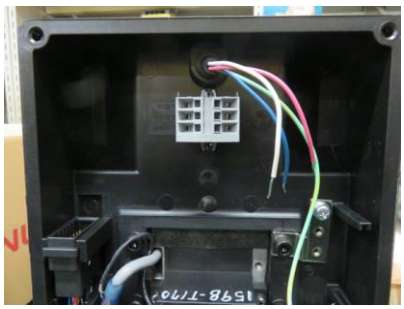


Terminal box of the motor

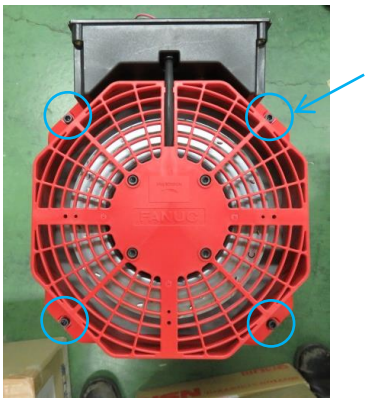
Inside the terminal box

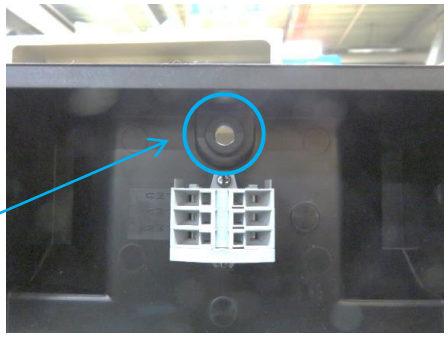
② Detach the power leads of fan motor from terminal block and ground wire from ground, both located inside the terminal box.





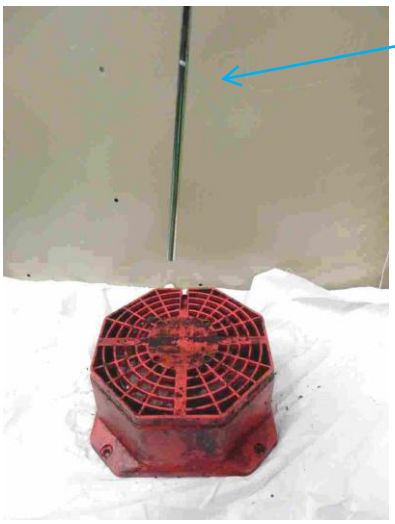
③ Unscrew the 4 bolts on the fan motor and detach the fan motor from the spindle motor. During this operation, be careful not to detach the rubber bushing from the terminal box hole.



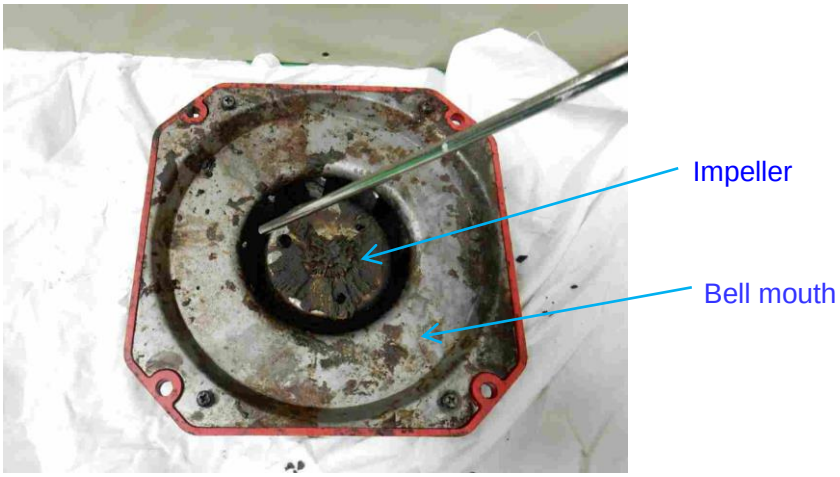


Cleaning of the Fan Motor

① Blow off the sludge accumulated around the fan motor by blowing compressed air. If the sludge cannot be blown off, please replace the fan motor.



② Blow compressed air from the space between bell mouth and impeller and blow off the sludge accumulated inside of the fan motor.



Blown off sludge

Reattaching the Fan Motor

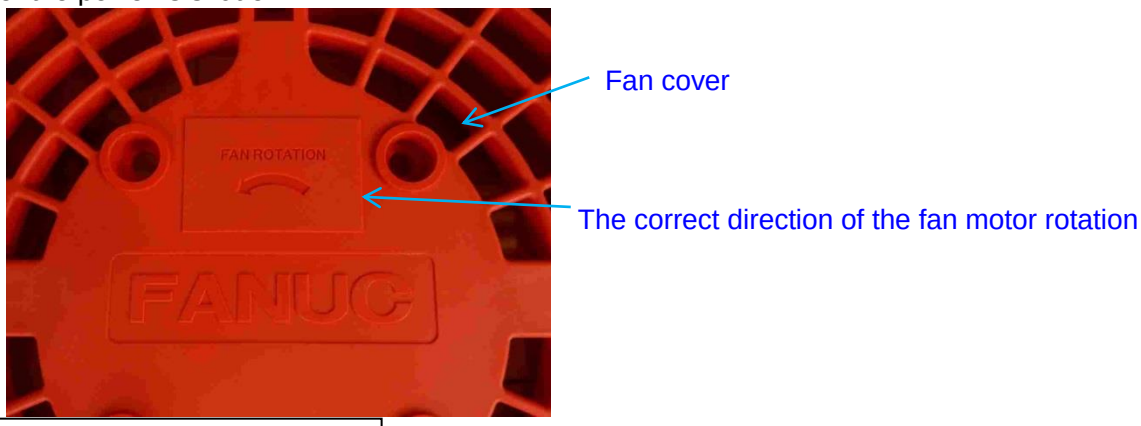
① Pull the power leads of the fan motor through the rubber bushing and draw them into the terminal box. Spraying lubricant on the rubber tube of the power leads will make the operation easier.
② Fix the fan motor on the spindle motor with 4 bolts.
On tightening the bolts, control the torque to the correct value. (Tightening torque: 2.8～4.0Nm)
③ Connect the power leads to the terminal block and ground wire to the ground in the terminal box.

Power leads wiring of the fan motor

The fan motor has 3 power leads and 1ground wire.
Wire the power leads to the terminals FMU, FMV and FMW printed on the terminal block in the terminal box or imprinted on the terminal box.
Wire the ground wire to the screw with the ground mark.

Type and color	Connection point
Power lead（Red）	terminal FMU
Power lead（White）	terminal FMV
Power lead（Blue or black）	terminal FMW
Ground wire（Green and yellow）	Screw of the ground terminal

④ Fix the lid on the terminal box with 4 bolts.
On tightening the bolts, control the torque to the correct value. (Tightening torque: 2.8～4.0Nm)
⑤ Verify that the fan motor is rotating in the correct direction by vision right after the power is put on or right after the power is shut off.



Replacement of the Fan Motor

① Detach the fan motor following the procedures in “Detaching the Fan Motor”.
② Have a replacement fan motor at hand and attach the new fan motor following the procedures in “Reattaching the Fan Motor”.

※ If a replacement fan is not available soon enough, make sure to shut off the power by dropping the breaker or disconnecting the power leads on the fan motor as a temporary treatment until the fan motor is replaced.